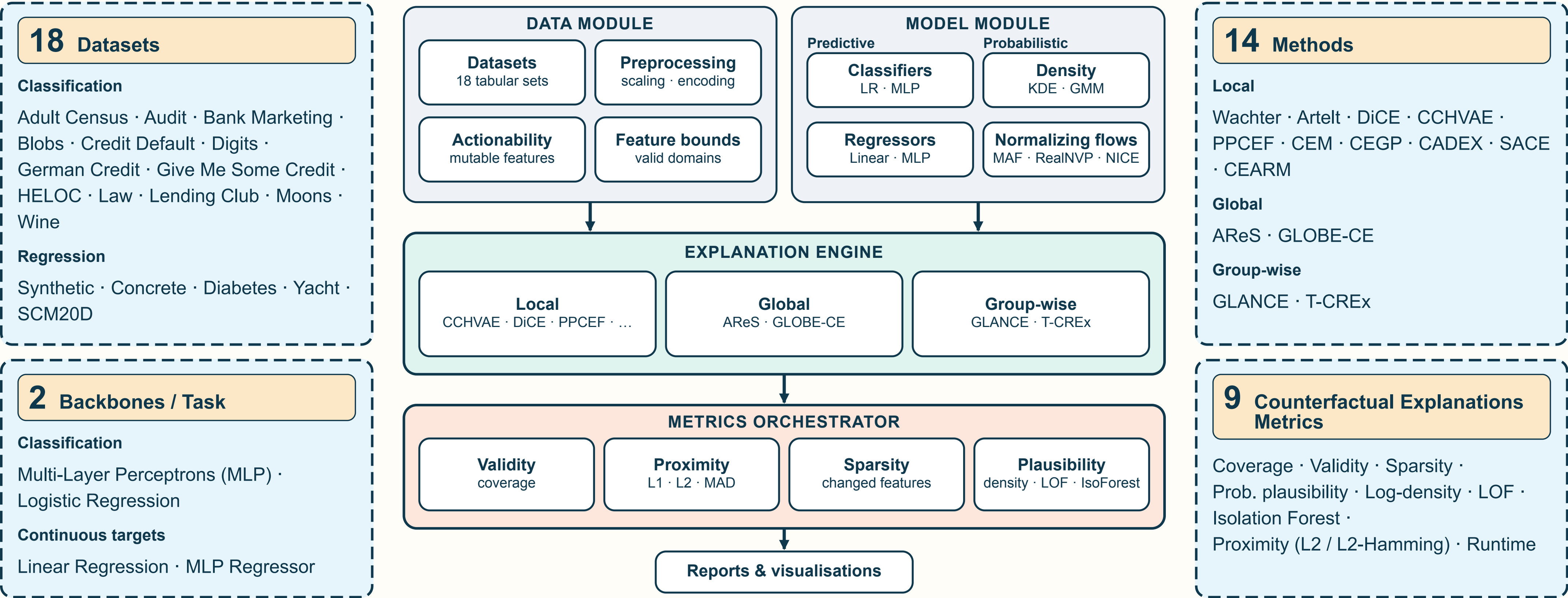




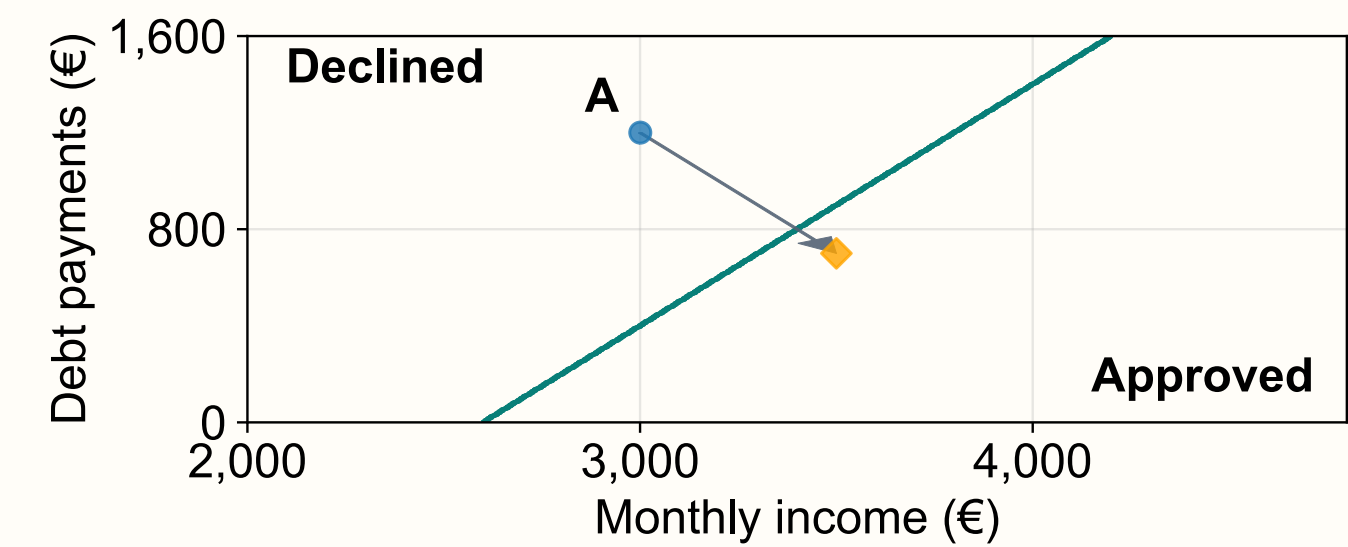
What is a counterfactual explanation?

A **counterfactual explanation** describes the smallest, realistic change in input data needed to flip an algorithmic decision.

It answers the “what-if” question by showing how a situation would look if specific facts were different. For example: “If your annual income was \$500 higher, your loan would have been approved.”

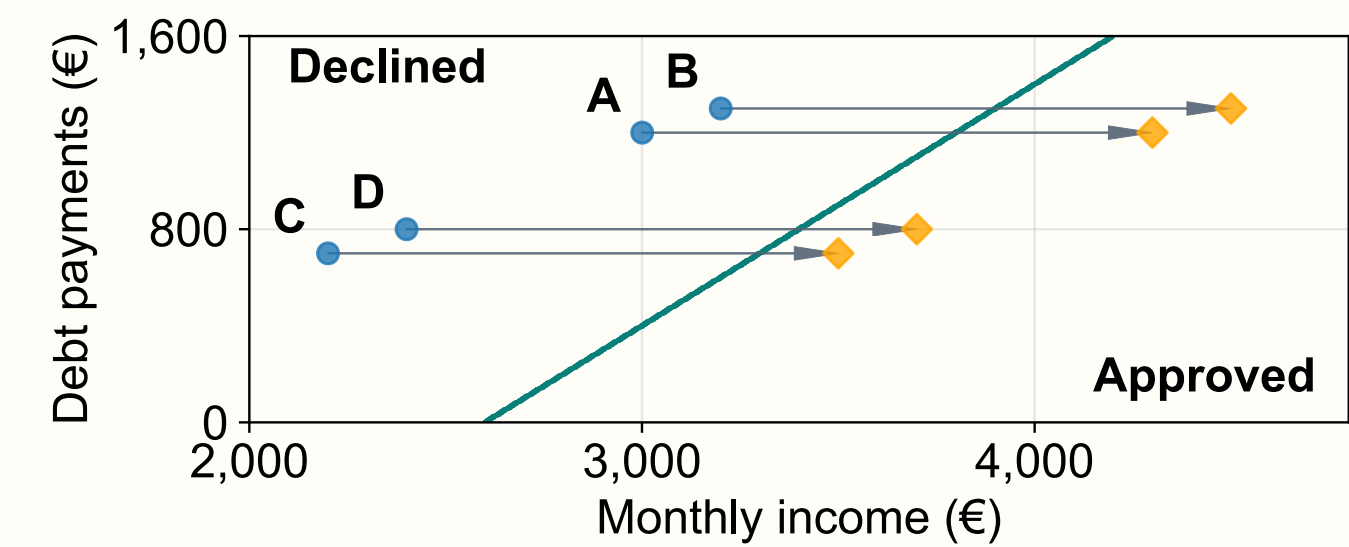
Local One applicant

A Income +€500; debt -€500.



Global One shared change

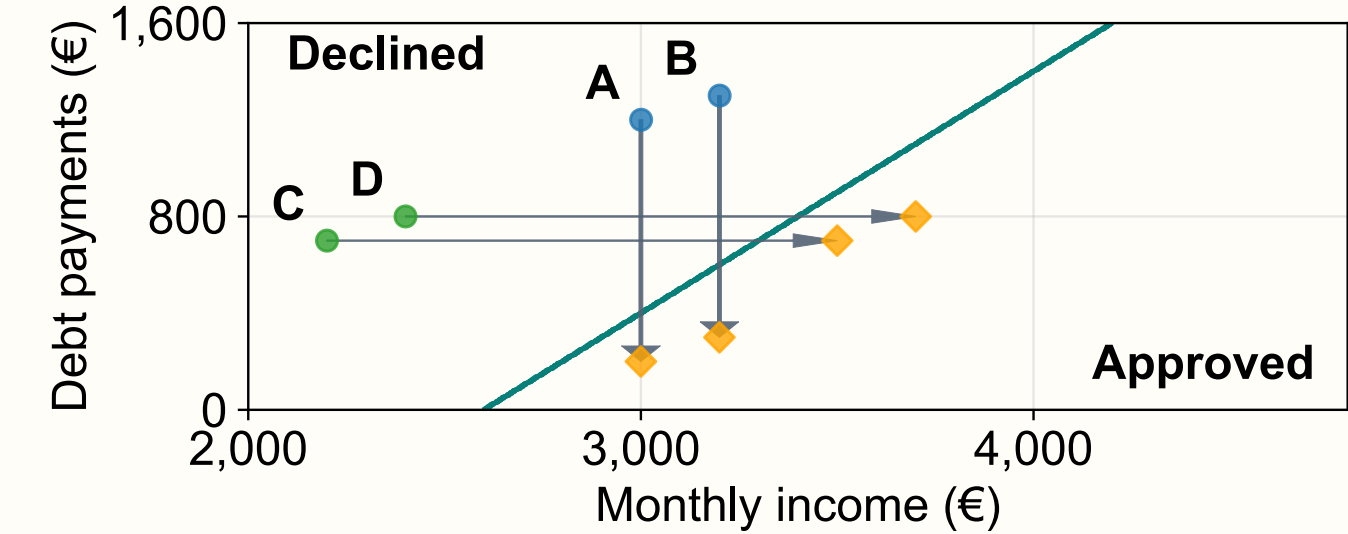
A-D Income +€1,300; debt unchanged.



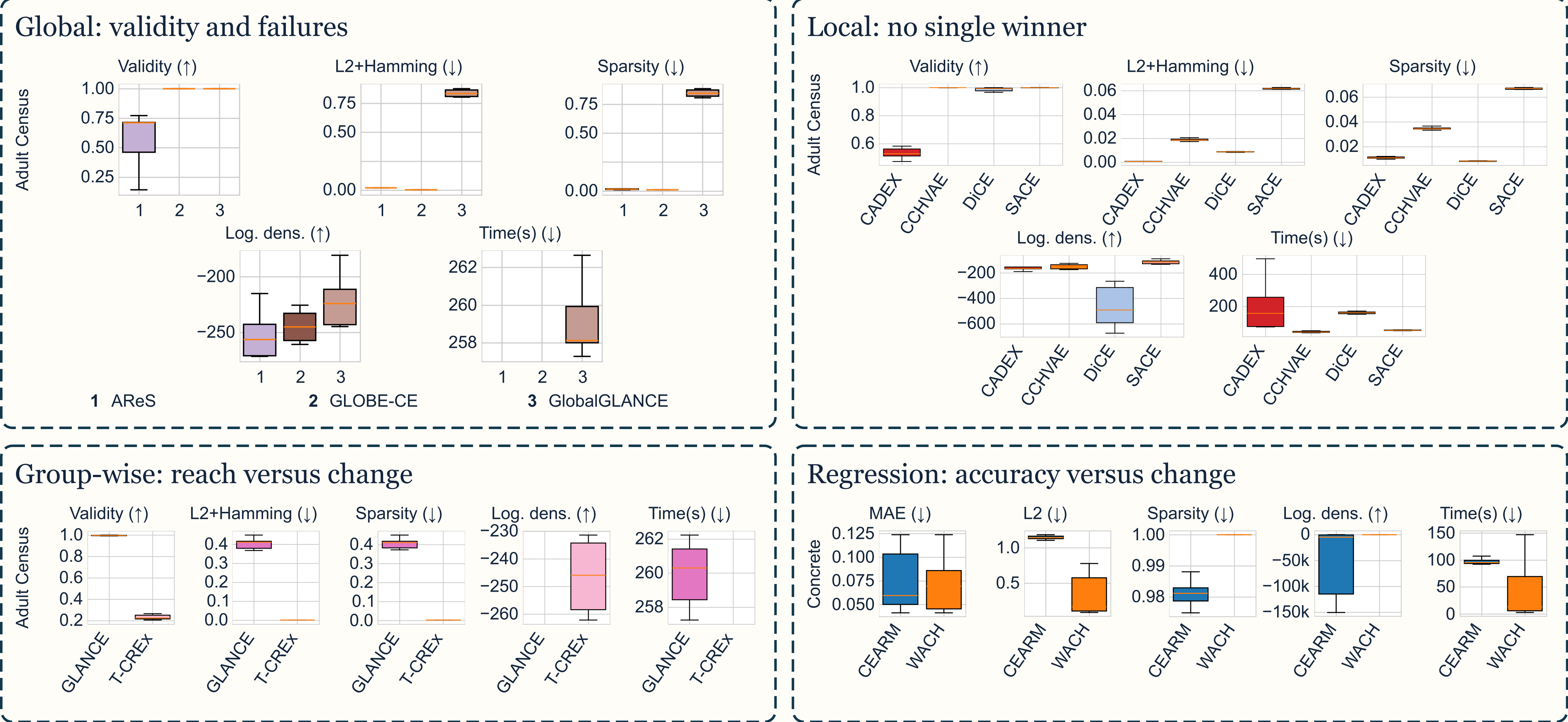
Group-wise One change per group

A-B Debt -€1,000; income unchanged.

C-D Income +€1,300; debt unchanged.



Results



Contributions

01 Controlled protocol

Reproducible evaluation

02 Cross-paradigm benchmark

Local · Global · Group-wise

03 Extensible library

Open source

